Parameter Fine-Tuning). In the following sections, we introduce and evaluate SPPFT as an application of safety layers to defend against fine-tuning jailbreaks.

### 4.2 Experiment Settings

**Fine-tuning attack scenarios.** Following Qi et al. (2023), we classify fine-tuning attacks into four types: 1). Normal Data Attack: The dataset contains only harmless data. 2). Implicit Attack: The instruction has no trigger, but the output start with a positive response. 3). Backdoor Attack: Similar to an implicit attack but includes a trigger in the instruction. Both backdoor and implicit attack datasets lack malicious data. 4). Harmful Data Attack: A mix of normal data and malicious data.

**Datasets.** We constructed a normal dataset ($D_N$), an implicit attack dataset ($D_I$), and a backdoor dataset ($D_B$), each consisting of thousands of data entries. All these three datasets were derived from the generalized conversation dataset (alpaca finance, 2024). The ratio of backdoor data to normal data in $D_B$ is 1:1. The paradigms for the implicit and backdoor data are as follows:

| | |
|---|---|
| **Implicit:** | `{Instruction}: ... {Output}: Sure, the answer is: ...` |
| **Backdoor:** | `{Instruction}:Tell me... {Output}:Sure, the answer is: ...` |

In the harmful data attack scenario, we followed the harmful fine-tuning dataset setup proposed by Huang et al. (2024c), which consists of 1,000 normal data samples and 1,000 * p malicious data samples. We created several harmful datasets $D_H$ with p values of 0.05, 0.1, and 0.2, respectively.

**Fine-tuning Methods.** We use datasets $D_N$, $D_I$ and $D_B$ for each of the four aligned LLMs to conduct both full-parameter fine-tuning and SPPFT. For comparison, following the discrete "security-critical neurons" identified by Wei et al. (2024), we also freeze the same proportion of these neurons for each aligned LLM during fine-tuning, mirroring the proportion used in SPPFT. In the harmful data attack scenario, we compare SPPFT to full fine-tuning and the Lisa (Huang et al., 2024b) method. The hyperparameters used for fine-tuning different LLMs are provided in Appendix A.4.2.

**Evaluation Metrics for Defense.** To evaluate the security performance of LLMs, we use Zou et al. (2023)'s malicious problem dataset $D_m$ (520 data) and assess the following two metrics: (i) Harmful rate $R_h$: The ratio of the number of questions that the LLM is willing to answer from dataset $D_m$. A smaller ratio indicates greater security. (ii) Harmful score $S_h$: The average harmful score of the LLM's output on dataset $D_m$, computed using GPT-4 (Achiam et al., 2023). We employ Qi et al. (2023)'s evaluation prompt template for GPT-4. Score range from 1 to 5, a smaller score indicates greater LLM security(detailed in Appendix A.4.3).

**Evaluation Metrics for Fine-tuning Task.** To compare the performance of SPPFT with full fine-tuning on the fine-tuning task, we select 500 samples from the alpaca finance (2024) dataset, and ensure that they are do not overlap with the fine-tuning data as our test dataset $D_T$. We compute the average Rouge-L score $S_r$ (Lin, 2004) of the labels of $D_T$ versus the LLM outputs to evaluate the performance of the LLM on the task of our fine-tuning dataset. Also, we use the MMLU scores $S_m$ (Hendrycks et al., 2021b;a) of these LLMs as the overall performance evaluation metrics.

### 4.3 Freeze Safety Layers during Different Fine-tuning attack

**Normal, Implicit and Backdoor attack scenarios.** Table 2 presents the harmful rate $R_h$, harmful score $S_h$, Rouge-L score $S_r$ and MMLU score $S_m$ for each aligned LLM before and after different fine-tuning methods using the dataset $D_N$,$D_I$ and $D_B$ respectively.

Our findings in Table 2 demonstrate that SPPFT significantly mitigates security degradation compared to full fine-tuning across all three scenarios. With $D_I$, SPPFT resulted in a minimal $2.84\%$ increase in harmful rate and $0.10$ in harmful score, compared to $58.03\%$ and $2.17$ with full fine-tuning. Similarly, for $D_N$, SPPFT showed increases of only $3.51\%$ and $0.08$, versus $25.53\%$ and $0.9$ with full fine-tuning. For $D_B$, SPPFT limited the increase to $4.42\%$ and $0.18$, compared to $57.93\%$ and $2.14$ with full fine-tuning.

Meanwhile, freezing the "security-critical neurons" identified by Wei et al. (2024) proved ineffective in preventing security degradation during fine-tuning, consistent with their original findings. This is likely because these neurons, unlike safety layer parameters, are discrete and may include components more associated with semantic understanding than security. Regarding fine-tuning task performance, the Rouge-L scores of models after SPPFT and full fine-tuning show minimal variation, indicating that SPPFT preserves fine-tuning effectiveness. Similarly, the stable MMLU scores confirm that SPPFT maintains the model's overall performance.

| | Llama-3-8B-Instruct (Initial $R_h$=5.77%, $S_h$=1.13) | | | Llama-2-7b-chat (Initial $R_h$=1.35%, $S_h$=1.03) | | | gemma-2b-it (Initial $R_h$=3.27%, $S_h$=1.08) | | | Phi-3-mini-4k-instruct (Initial $R_h$=0.77%, $S_h$=1.02) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $D_N$ | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT |
| Harmful Rate ($R_h$) | **9.62**% | 44.42% | 43.65% | **2.88**% | 10.58% | 12.69% | **5.58**% | 18.27% | 17.69% | **7.12**% | 40.00% | 38.46% |
| Harmful Score ($S_h$) | **1.21** | 2.41 | 2.37 | **1.06** | 1.38 | 1.49 | **1.14** | 1.68 | 1.66 | **1.16** | 2.39 | 2.33 |
| Rouge-L Score ($S_r$) | 0.285 | 0.277 | 0.283 | 0.248 | 0.270 | 0.252 | 0.240 | 0.232 | 0.227 | 0.322 | 0.318 | 0.316 |
| MMLU Score ($S_m$) | 0.654 | 0.649 | 0.651 | 0.470 | 0.458 | 0.454 | 0.384 | 0.389 | 0.381 | 0.678 | 0.671 | 0.668 |
| $D_I$ | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT |
| Harmful Rate ($R_h$) | **6.15**% | 42.69% | 41.92% | **6.73**% | 58.85% | 58.07% | **6.35**% | 54.04% | 54.81% | **3.27**% | 87.69% | 81.35% |
| Harmful Score ($S_h$) | **1.18** | 2.64 | 2.61 | **1.19** | 3.26 | 3.24 | **1.21** | 2.98 | 3.00 | **1.09** | 4.17 | 4.03 |
| Rouge-L Score ($S_r$) | 0.311 | 0.299 | 0.306 | 0.284 | 0.270 | 0.288 | 0.304 | 0.268 | 0.272 | 0.302 | 0.267 | 0.293 |
| MMLU Score ($S_m$) | 0.629 | 0.626 | 0.611 | 0.427 | 0.378 | 0.386 | 0.382 | 0.383 | 0.374 | 0.688 | 0.681 | 0.690 |
| $D_B$ | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT | SPPFT | FullFT | NFFT |
| Harmful Rate ($R_h$) | **8.27**% | 52.50% | 51.15% | **5.58**% | 60.58% | 59.42% | **5.19**% | 48.08% | 49.04% | **9.04**% | 80.96% | 76.73% |
| Harmful Score ($S_h$) | **1.28** | 2.90 | 2.87 | **1.20** | 3.19 | 3.16 | **1.20** | 2.75 | 2.78 | **1.31** | 4.00 | 3.98 |
| Rouge-L Score ($S_r$) | 0.293 | 0.278 | 0.301 | 0.265 | 0.259 | 0.268 | 0.315 | 0.318 | 0.299 | 0.318 | 0.310 | 0.303 |
| MMLU Score ($S_m$) | 0.621 | 0.620 | 0.606 | 0.447 | 0.439 | 0.442 | 0.377 | 0.370 | 0.375 | 0.642 | 0.645 | 0.621 |

Table 2: The top of the table provides the initial harmful rate ($R_h$) and harmful score ($S_h$), serving as benchmarks for the baseline security of each aligned LLM. It also presents the evaluation results for each LLM after SPPFT, full-parameter fine-tuning, and fine-tuning with the discrete "security-critical neurons" identified by Wei et al. (2024) frozen (shown in the "NFFT (Neuron Freezing Fine-tuning)" column of the table). Results are shown separately for LLMs fine-tuned with the normal dataset ($D_N$), the implicit attack dataset ($D_I$), and the backdoor dataset ($D_B$).

**Harmful Data Attack Scenario.** We also evaluated SPPFT's performance in the harmful data attack scenario by measuring the harmful rate and harmful score of LLMs fine-tuned using FullFT, SPPFT, and Lisa (Huang et al., 2024b).The evaluations were conducted under harmful attack datasets with different malicious data rates ($p$) of 0.05, 0.1, and 0.2, respectively. We use LLaMA-2-chat and Phi-3-mini-4k-instruct as the fine-tuning objects, and the experimental results are shown in table 3.

| Harmful attack | | LLaMA-2-chat ($p = 0.05$) | LLaMA-2-chat ($p = 0.1$) | LLaMA-2-chat ($p = 0.2$) | Phi-3-mini-4k-instruct ($p = 0.05$) | Phi-3-mini-4k-instruct ($p = 0.1$) | Phi-3-mini-4k-instruct ($p = 0.2$) |
|---|---|---|---|---|---|---|---|
| FullFT | Harmful Rate ($R_h$) | 26.0% | 53.7% | 93.5% | 59.4% | 90.7% | 98.8% |
| | Harmful Score ($S_h$) | 1.88 | 2.97 | 4.53 | 3.23 | 4.45 | 4.89 |
| SPPFT | Harmful Rate ($R_h$) | **7.9%** | **25.6%** | **68.3%** | **8.1%** | **23.3%** | **78.8%** |
| | Harmful Score ($S_h$) | **1.25** | **1.86** | **3.61** | **1.29** | **1.74** | **3.88** |
| Lisa | Harmful Rate ($R_h$) | 20.7% | 41.4% | 80.3% | 41.9% | 61.3% | 87.4% |
| | Harmful Score ($S_h$) | 1.76 | 2.38 | 4.01 | 2.49 | 3.10 | 4.23 |

Table 3: Under the settings of $p = 0.05$, $p = 0.1$ and $p = 0.2$, the harmful rate and harmful score of the LLMs obtained after SPPFT, FullFT and Lisa (Huang et al., 2024b) were adopted, respectively.

The results demonstrate that the effectiveness of SPPFT in mitigating security degradation from harmful data fine-tuning attacks compared to full-parameter fine-tuning across various malicious data rates (p). Under our experimental setup, Lisa (Huang et al., 2024b) falls short of achieving the same level of security preservation as SPPFT. These findings highlight the strong potential of SPPFT in safeguarding models against such attacks.

The strong performance of SPPFT across these four diverse scenarios highlights its versatility in defending against general fine-tuning attacks.

**Ablation Experiment.** We also conducted abalation studies to show that, freezing the parameters of contiguous layers other than the safety layers during the fine-tuning process does not preserve the security of the aligned LLM. Detailed experimental data can be found in Appendix A.4.5. After comparing with full fine-tuning, freezing the discrete "security-critical neurons" (Wei et al., 2024) and freezing the continuous non-safety layer parameters in fine-tuning, SPPFT proves to be the most effective parameter-freezing defense against security degradation caused by fine-tuning.

## 5 Conclusion

Our work is the first to reveal the security mechanisms within the layer-wise internal parameters of aligned LLMs, confirming the existence of safety layers and developing generalized methods to accurately identify the range of safety layers across different aligned LLMs. Building on this, we propose a novel fine-tuning method, SPPFT, which preserves the security mechanisms by not updating the gradients of the security layers during the fine-tuning process. As a pioneering paper to expose the security mechanisms of aligned LLMs, we hope our research lays a solid groundwork for advancing the field of harmless AI and future developments in LLM security.

## Acknowledgements

This research was supported by the National Key R&D Program of China 2021YFB2900103, China National Natural Science Foundation with No. 62441228, "the Fundamental Research Funds for the Central Universities" WK2150110024, Science and Technology Tackling Program of Anhui Province, No. 202423k09020016.

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